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SE Lab – 3

1. Architecture Selection

We chose Layered Architecture for the Telemedicine Slot Booking & Prescription Portal.

2. Architectural Choice

Layered Architecture is selected because it provides a clear separation between the presentation, business logic, and data layers of the telemedicine system. This structure is suitable for organizing the portal's doctor discovery, consultation booking, teleconsultation, and prescription management functions.

i) Reason 1 — Clear Separation of Functionalities

The system contains distinct functionalities such as viewing doctors, booking consultation slots, generating secure consultation links, and managing digital prescriptions. A layered architecture separates these responsibilities into appropriate layers, making the system easier to maintain and manage.

ii) Reason 2 — Suitable for Secure Data Management

The portal handles sensitive patient clinical notes, consultation information, and digitally signed prescriptions. A layered structure allows security and authorization rules to be enforced before sensitive data is accessed or modified. This supports the security requirements defined for the system, including TLS 1.3 and AES-256 encryption.

3. Security Advantage

Layered Architecture allows authentication, authorization, and secure data-access controls to be centralized within the appropriate application and data layers. This helps restrict sensitive patient and consultation information to authorized users and supports the required encryption mechanisms.

4. Performance Benefit

The separation of presentation, business logic, and data access allows the consultation-booking process to be optimized independently within the business layer. This can help the system provide confirmed appointment details and encrypted consultation links within the required 5-second response target.